

MRCPsych Paper B

Recall Trend Analysis & Study Guide

Based on Recall Papers: Sep 2022 | Mar 2023 | Jun 2023 | Oct 2023 | Mar 2024 | May 2025

6 exam sessions • SPMM Mock 14 (2024) • Covers every major examinable domain

SECTION 1: EXAM STRUCTURE & OVERALL TRENDS

After analysing all 8 source files (6 recall papers spanning Sep 2022 – May 2025, SPMM Mock 14, and a compiled solved recall), the following picture emerges:

Broad Domain	Approx. % of Paper	Trend
Statistics & Research Methods	30–35%	CONSISTENT – every session, high volume
Clinical Pharmacology (inc. pregnancy, side effects)	20–25%	CONSISTENT – detailed & calculation-heavy
Intellectual Disability & Genetics	8–12%	CONSISTENT – same syndromes recycled
Psychotherapy	8–10%	CONSISTENT – specific key terms tested
Dementia & Old Age	8–10%	CONSISTENT – mixed pharmacology + stats
Forensic Psychiatry	6–8%	CONSISTENT – specific facts tested
Child & Adolescent Psychiatry	6–8%	CONSISTENT – NICE guidelines focus
Eating Disorders & Perinatal	4–6%	CONSISTENT
Prevention	2–3%	CONSISTENT – same question variants

CRITICAL OBSERVATION: The same questions appear across multiple sessions with only minor wording changes. Learning the pattern is as important as learning the content.

SECTION 2: STATISTICS & RESEARCH METHODS – THE HIGHEST-YIELD TOPIC

Statistics makes up ~30–35% of Paper B. It appears in every session with consistent subtopics. This is the single most important area to master.

2.1 Diagnostic Test Statistics (appears EVERY session, multiple questions)

Concept	Formula / Key Point	Exam Trap
Sensitivity	$TP / (TP + FN)$	Probability of +ve test in diseased
Specificity	$TN / (TN + FP)$	Probability of –ve test in non-diseased
PPV	$TP / (TP + FP)$	Chance patient HAS disease if test +ve
NPV	$TN / (TN + FN)$	Chance patient does NOT have disease if test –ve
LR+	$Sensitivity / (1 - Specificity)$	How much odds increase with +ve result
LR–	$(1 - Sensitivity) / Specificity$	How much odds decrease with –ve result
Post-test odds	$Pre-test\ odds \times LR+$	Calculate step by step (1 in 8 $\rightarrow$ 1/7)
ARR / NNT	$CER - EER$; $NNT = 1/ARR$	Read forest plot for event rates first

Exam favourite: Pre-test prob 1/8, Sn 80%, Sp 80% $\rightarrow$ Pre-test odds $1/7 \times LR+ 4 =$ post-test odds $4/7$. This appeared in Sep 2022, Mar 2023, Mock 14.

If sensitivity INCREASES (cut-off unchanged): false positives INCREASE, specificity DECREASES. PPV increases, NPV stable, power increases.

If CAGE cut-off raised from 2 to 3: sensitivity DECREASES, specificity INCREASES.

2.2 Statistical Tests (selection tested in every paper)

Situation	Correct Test
Comparing 2 groups, continuous normal data, different patients	Unpaired (independent) t-test
Comparing 2 groups, same patients before/after	Paired t-test
Comparing 3+ groups, continuous normal data	One-way ANOVA

Situation	Correct Test
ANOVA adjusting for a covariate (e.g. age)	ANCOVA
Comparing 2 groups, non-normal/ordinal	Mann-Whitney U
Comparing 3+ groups, non-normal/ordinal	Kruskal-Wallis
Paired non-parametric data	Wilcoxon signed-rank
2 categorical variables (2x2 table, $n \geq 5$)	Chi-square
2 categorical variables (expected cell < 5)	Fisher's exact test
Repeated categorical measures on SAME patients	McNemar test
Continuous outcome, multiple predictors	Multiple linear regression
Dichotomous outcome, multiple predictors	Logistic regression
Relationship between 2 continuous normal variables	Pearson correlation
Relationship between 2 ordinal/non-normal variables	Spearman correlation
Convert skewed data to normal distribution	Log transformation
Survival analysis over time	Kaplan-Meier / Cox regression
Association between BP and height, controlling for weight	Partial correlation / ANCOVA

Frequently tested trap: ANCOVA vs ANOVA. ANCOVA is used when you need to adjust for a covariate (e.g. age). This appeared in Oct 2023, Jun 2023, May 2025.

2.3 Study Design (appears every session)

Study Type	Key Feature	When to Choose
RCT	Random allocation, gold standard for efficacy	Testing if a treatment works
Cluster RCT	GP surgeries/schools randomised, not individuals	Service-level interventions
Cohort	Follow exposed vs unexposed forward in time	Incidence, RR calculation, rare exposures
Case-control	Cases vs controls, work backward	Rare disease, OR calculation, quick/cheap
Nested case-control	Case-control within a cohort	Antipsychotic + hyperglycaemia scenario
Cross-sectional	Snapshot, prevalence only	Survey, screening tool validation
Ecological	Population-level data (e.g. lithium in water)	Cannot make individual inferences
Qualitative	Explores experiences, not numbers	Grounded theory, phenomenology, thematic analysis

Study Type	Key Feature	When to Choose
N of 1	Single patient crossover trial	Initial testing of a new drug on one person
Service evaluation	Auditing against standards (not research)	Assessing NHS service performance

Case-control → odds ratio (OR). Cohort → relative risk (RR). Risk ratio CANNOT be calculated in case-control (no denominator from population).

2.4 Bias, Randomisation & Analysis

- Block randomisation → ensures EQUAL numbers in each arm (most tested randomisation question)
- Stratification → controls for confounders at the ANALYSIS stage (also controls for effect modifiers)
- Minimisation → next patient allocated to whichever group has FEWER of their characteristics already
- Cluster randomisation → whole GP surgeries or schools randomised together
- Intention-to-treat (ITT) → controls attrition bias; everyone analysed as randomised regardless of compliance
- Per-protocol analysis → only compliant patients; used for biological efficacy studies
- Multiple imputation → fills missing values from observed data distributions
- Blinding → controls measurement bias; allocation concealment → controls selection bias
- Fixed effects model → assumes error is WITHIN studies only (no heterogeneity between studies)
- Random effects model → used when significant HETEROGENEITY exists between studies
- I^2 statistic → measures heterogeneity; >50% = substantial; >75% = considerable

Audit trap: If a re-audit is done but no changes were implemented in between → the audit loop is incomplete. This appeared in Sep 2022, Oct 2023, Recalls Solved.

2.5 Validity & Reliability

Term	Definition	Example from Recalls
Concurrent validity	New test compared to gold standard at SAME time	New depression scale vs Hamilton, both given together
Predictive validity	Score predicts a FUTURE outcome	Paper B score predicts CASC result; suicide scale predicts self-harm
Convergent validity	Two measures of SAME construct correlate	Paper A vs Paper B score correlation
Divergent validity	Test distinguishes DIFFERENT constructs	ASD scale gives +ve for ASD but not ADHD
Content validity	Scale covers ALL aspects of the construct	Depression scale missing somatic symptoms = lacks content validity

Term	Definition	Example from Recalls
Face validity	Looks like it tests what it should	Snapshot / surface appearance only
Inter-rater reliability	Same result from 2 different raters	2 radiologists report same hippocampal atrophy
Internal consistency	Items measure the SAME thing	Cronbach's alpha; split-half reliability
Test-retest reliability	Same results on repeat testing	Stable measures over time

2.6 Qualitative Research (appears every session, 3-5 questions)

- Bracketing (phenomenology) → researcher sets aside preconceptions before analysis
- Member checking / respondent validation → comparing researcher's account with participants' views
- Triangulation → using multiple data sources or methods to reduce bias
- Reflexivity → researcher acknowledges and reflects on their own influence on the research
- Data saturation / theoretical saturation → no new themes emerging = adequate sample size
- Purposive sampling → deliberately selecting participants with specific traits
- Grounded theory → theory generated FROM data (bottom-up)
- Thematic analysis → identifying themes across qualitative data
- Content analysis → systematic analysis of text/media for patterns (e.g. newspaper articles)
- Ethnography → studying cultural groups through immersion and observation

2.7 Health Economics (appears every session)

Term	Key Definition
Cost-utility analysis	Uses QALYs as outcome; $ICER = (cost\ A - cost\ B) / (QALY\ A - QALY\ B)$
Cost-effectiveness analysis	Single clinical outcome (e.g. depression-free days); ICER reported
Cost-benefit analysis	All outcomes expressed in MONETARY terms
Cost-minimisation analysis	Both treatments equivalent; choose CHEAPEST
Direct costs	Hospital, staff, medications, transport to appointment
Indirect costs	Lost productivity, sick leave, carer time
Intangible costs	Pain, suffering, stigma — cannot be easily quantified
Opportunity cost	Benefit forgone by using resource elsewhere (e.g. nurse moved from pain clinic to ECT)
Incremental cost	Additional cost of one treatment over another
QALY calculation	Life years $\times$ quality weight (0–1); 40 years $\times$ 0.7 = 28 QALYs

2.8 Reporting Standards (STARD appeared Mar 2023, May 2025)

Guideline	For what type of study
CONSORT	Randomised controlled trials
PRISMA	Systematic reviews and meta-analyses
STARD	Diagnostic accuracy studies (sensitivity/specificity)
STROBE	Observational studies in epidemiology
QUOROM	Meta-analyses of RCTs (superseded by PRISMA)
MOOSE	Meta-analyses of observational studies
COREQ	Qualitative research
SQUIRE	Quality improvement studies

SECTION 3: INTELLECTUAL DISABILITY & GENETICS

These questions appear in every paper. The SAME syndromes are tested repeatedly — learn the distinguishing features cold.

3.1 LD Syndrome Comparison Table

Syndrome	Genetics	Key Clinical Features	Exam Clue
Fragile X	X-linked dominant; FMR1 gene; CGG repeat	Gaze avoidance, cluttered speech, hyperactivity, anxiety, macroorchidism, large ears	Jocular speech + gaze aversion
Down syndrome	Trisomy 21 (95%); Robertsonian translocation (5%, inherited); Mosaic	APOE4 risk for Alzheimer's by age 40; App gene on chr 21; passive/affable behaviour	Robertsonian = inherited equally from either parent
Lesch-Nyhan	X-linked recessive; HPRT gene; hyperuricaemia	Severe self-mutilation (biting lips/fingers), spasticity, choreiform movements, microcephaly	Self-biting + choreiform + microcephaly in boy
PKU	Autosomal recessive; PAH deficiency	Mouse-like odour, hypopigmentation, microcephaly, hyperactivity, severe LD, seizures	Fair skin + mouse odour + microcephaly

Syndrome	Genetics	Key Clinical Features	Exam Clue
Smith-Magenis	Del 17p11.2	Self-hugging posturing, sleep disturbance, self-injury, hoarse voice, flat midface	Self-hugging is pathognomonic
Cri du chat	Del 5p; karyotype 5p-	Cat-like cry, hypertelorism, round face, hypotonia, severe LD	High-pitched cat cry + hypertelorism
Rett syndrome	MECP2 gene (X-linked); affects girls only	Normal development then regression at 6–18 months, hand-wringing, epilepsy, breathing difficulties	Girl + regression + hand wringing
Angelman	Maternal deletion 15q; genomic imprinting	Happy puppet, severe LD, paroxysmal laughter, no speech, hand-flapping, epilepsy (86%)	Happy puppet + loves water/music
Prader-Willi	Paternal deletion 15q; genomic imprinting	Hyperphagia, obesity, hypotonia, 1:1 M:F, tantrums, skin-picking	Male = female prevalence; genomic imprinting
DiGeorge / 22q11.2	Chr 22 deletion; CATCH-22	Psychosis in 20–30%, cardiac abnormalities, cleft palate, hypocalcaemia, hypertelorism	Psychosis + cardiac + hypocalcaemia
Tay-Sachs	Autosomal recessive; beta-hexosaminidase A deficiency	Ashkenazi Jews, cherry-red spot, progressive neurodegeneration	Ashkenazi + cherry-red spot
Friedreich's ataxia	Autosomal recessive; GAA repeat on chr 9; frataxin	Gait ataxia, falls, nystagmus, cardiac disease (leading cause of death), MI in grandfather	Young man + ataxia + grandfather died from MI
Williams syndrome	Del 7q11.23	Friendly/disinhibited, anxiety, verbal > visuospatial, cardiovascular abnormalities	Overfriendly + excellent verbal skills
Huntington's	Autosomal dominant; CAG repeat (>36 =	Chorea, MRI shows caudate atrophy, onset 30–50s	CAG >36 + MRI

Syndrome	Genetics	Key Clinical Features	Exam Clue
	disease); chromosome 4		caudate atrophy

Genomic imprinting: Prader-Willi = paternal deletion 15q. Angelman = maternal deletion 15q. Fragile X = error in imprinting. This is tested in Mar 2024 and Recalls Solved.

3.2 Inheritance Patterns

- Autosomal dominant: Huntington's, Neurofibromatosis 1&2, tuberous sclerosis, Noonan's
- Autosomal recessive: PKU, homocystinuria, Tay-Sachs, Wilson's disease, Hurler's, Friedrich's ataxia, cystic fibrosis
- X-linked recessive: Lesch-Nyhan, Hunter syndrome, Fragile X (technically X-linked dominant)
- X-linked dominant: Rett syndrome (MECP2), Fragile X

3.3 Causes of Intellectual Disability

- Most common OVERALL cause → Unknown (up to 60%)
- Most common CHROMOSOMAL cause → Down syndrome (Trisomy 21)
- Most common INHERITED/GENETIC cause → Fragile X
- Most common ENVIRONMENTAL/ACQUIRED cause → Foetal alcohol syndrome

SECTION 4: DRUGS IN PREGNANCY & NEONATAL EFFECTS

This appears as an EMI in every paper. Learn the neonatal presentations for each drug and the teratogenic risks.

4.1 Teratogenic Effects of Psychiatric Drugs

Drug	Teratogenic Risk	Neonatal Effect
Sodium valproate	Cleft palate, neural tube defects (spina bifida 1–2%), cardiac, limb abnormalities, developmental delay, autism	Irritability, developmental delay
Lithium	Ebstein's anomaly (cardiac, not as high risk as previously thought)	Floppy baby, cyanosis, bradycardia; suspend 24–48h before delivery; restart day 1 postnatal; check level 12h after last dose, then at 1 week

Drug	Teratogenic Risk	Neonatal Effect
Paroxetine	VSD/ASD (cardiac malformations) especially in 1st trimester; highest cardiac risk of SSRIs	Irritability, neonatal discontinuation syndrome, PPHN
SSRIs (late pregnancy)	Persistent pulmonary hypertension of newborn (PPHN) – absolute risk very low	Mild irritability / neonatal discontinuation; PPHN
Carbamazepine	Neural tube defects, cleft palate, cardiac	Vitamin K deficiency
Quetiapine	Preferred antipsychotic in pregnancy (low placental passage)	Relatively safe
Sertraline	Preferred SSRI in pregnancy (least placental exposure) and breastfeeding	Mild discontinuation

Safest antidepressant in breastfeeding = sertraline (and paroxetine has low milk/plasma ratio). Lithium = CONTRAINDICATED in breastfeeding.

If the question involves tamoxifen → AVOID paroxetine and fluoxetine (potent CYP2D6 inhibitors that reduce endoxifen). Safe alternatives: sertraline, citalopram, venlafaxine.

4.2 Neonatal Drug Effects (classic EMI)

Neonatal Presentation	Most Likely Maternal Drug
Jittery, high-pitched cry, poor feeding, sleep-wake abnormalities, vomiting	Opioids (heroin)
Overarousal, tachypnoea, tachycardia, agitation	Cocaine
Mild tremors, intermittent irritability, increased need for handling, normal reflexes	Nicotine
Floppy baby, breathing difficulties, hypotonic muscles, irritability	Alcohol (foetal alcohol syndrome: also microcephaly, dysmorphic facies, 2kg birth weight)
Microcephaly, dysmorphic facial features, low birth weight	Alcohol (FAS)
Cleft palate, developmental delay, cardiac malformations	Sodium valproate
Jitteriness 24 hours after birth	Heroin / opioids

SECTION 5: DEMENTIA & OLD AGE PSYCHIATRY

5.1 Dementia Types – Key Distinguishing Features

Type	Key Features	Imaging	Treatment
Alzheimer's	Short-term memory first; temporoparietal dysfunction; APOE4 allele; risk doubles every 5 years after 65; ×16 at age 85 vs 65	CT: medial temporal/hippocampal atrophy; SPECT: temporal & parietal hypoperfusion	Donepezil (reduce dose for GI SEs); memantine for moderate-severe
Dementia with Lewy bodies (DLB)	CORE: REM sleep behaviour disorder; Parkinsonism; visual hallucinations; fluctuating cognition. SUPPORTIVE: neuroleptic sensitivity (severe), falls, syncope	CT/MRI: medial temporal lobes SPARED; DAT scan: reduced dopamine	Rivastigmine (best evidence); AVOID antipsychotics
Vascular	Stepwise decline; greatest risk factor = HYPERTENSION; Binswanger's = small vessel disease, white matter changes	MRI: lacunar infarcts, white matter hyperintensities	Treat vascular risk factors
Frontotemporal (FTD)	Behaviour first (bvFTD); disinhibition, apathy, hyperphagia. Language variants: nvPPA = non-fluent aphasia; svPPA = semantic problems. Genes: C9orf72, GRN, MAPT	Frontal/temporal atrophy	No licensed pharmacotherapy
Parkinson's disease dementia	Parkinsonian symptoms >12 months BEFORE dementia. Visual hallucinations. If <12 months → consider DLB	Dopaminergic deficit	Rivastigmine; quetiapine/clozapine for psychosis
Normal pressure hydrocephalus	Triad: gait (magnetic), urinary incontinence, dementia. REVERSIBLE with shunt	CT/MRI: ventricular dilation, thin cortex	VP shunt
MND dementia	Only IRREVERSIBLE cause of dementia in the list. ALS + FTD overlap	—	No treatment modifies disease

Frequently tested: Which dementia is IRREVERSIBLE? MND/ALS. Reversible causes include NPH, chronic subdural haematoma, Wilson's disease, pellagra (niacin deficiency), alcohol.

DLB – REM sleep behaviour disorder is a CORE feature that can precede cognitive decline by years.
Supportive feature = neuroleptic sensitivity.

5.2 MCI vs Dementia

- MCI = middle ground between normal ageing and dementia; ADLs are PRESERVED in MCI

- Best tool to differentiate MCI from dementia = INFORMANT assessment of activities of daily living (not MMSE or MoCA alone)
- MoCA is more sensitive than MMSE for MCI detection
- NART (National Adult Reading Test) = measures PREMORBID intelligence
- RUDAS = cognitive assessment for non-English speakers / low education

5.3 BPSD Management in Dementia

- FIRST step → non-pharmacological interventions always
- Risperidone = ONLY licensed antipsychotic for BPSD in the UK (moderate-severe Alzheimer's with risk of harm)
- Haloperidol = not first-line; acceptable for severe acute agitation
- Avoid antipsychotics in DLB – risk of severe neuroleptic sensitivity
- SSRIs (sertraline, citalopram) = evidence for agitation in dementia; AVOID TCAs
- If patient has acute brain infarct → AVOID psychotropics if at all possible

SECTION 6: SUBSTANCE MISUSE

6.1 Drug Effects, Withdrawal & Key Features

Drug	Intoxication	Withdrawal	Special Points
Opioids	Euphoria, drowsiness, pupillary constriction, respiratory depression	Piloerection, insomnia, dilated pupils, yawning, sweating, abdominal cramps (not life-threatening)	Naloxone for overdose; methadone/buprenorphine for detox; naltrexone after abstinence (wait 7–10 days)
Cocaine/amphetamines	Tachycardia, hypertension, anorexia, paranoia, formication, euphoria	Vivid dreams, hunger (hyperphagia), fatigue, dysphoria / crash, depression, agitation	Cocaine = sustained attention most impaired long-term; amphetamine testing 48 hours in urine
Alcohol	Disinhibition, ataxia, slurred speech, blackouts	Starts 3–12 hours; peaks 24–48 hours; DTs at 3–4 days; LIFE-THREATENING	Oxazepam/lorazepam for liver disease; Wernicke = thiamine IV FIRST; acamprosate for abstinence; disulfiram CI in cardiac failure
Cannabis	Mild euphoria, giggling, relaxation, injected conjunctivae, tachycardia	Insomnia, reduced appetite, irritability (not life-threatening)	Anandamide = endogenous ligand; CB1/CB2 receptors; FEP risk related to FREQUENCY of use; least dependence = LSD

Drug	Intoxication	Withdrawal	Special Points
Ketamine	Dissociation, hallucinations, nystagmus, muscle rigidity, analgesia	No recognised withdrawal syndrome	Urinary symptoms (cystitis, haematuria) in 30% of users; Class B (UK); acts on NMDA and GABA-A
MDMA/Ecstasy	Jaw clenching, enhanced sociability, increased response to touch, euphoria, hyperthermia	Depression, insomnia, depersonalisation/derealisation	Risk of hyponatraemia from excessive water intake
LSD	Perceptual changes, tachycardia, sweating, pupillary dilation	NO recognised withdrawal; NO physical dependence	Least likely to cause dependence of all illicit drugs
Mephedrone	Stimulant; synthetic cathinone; Class B	Stimulant effects	Can give false positive for amphetamines

Alcohol and benzodiazepines are the **ONLY** drugs with life-threatening withdrawal. Opioid withdrawal is very unpleasant but not life-threatening.

GABA-A agonist + NMDA antagonist = **ALCOHOL** (both). Ketamine = NMDA antagonist only.

6.2 Alcohol Withdrawal Management

- Chlordiazepoxide = standard first-line for alcohol detox
- Oxazepam = choice in LIVER disease (phase 2 metabolism only) and COPD
- Lorazepam = choice when IV/IM route needed in liver disease
- Acamprosate = licensed for RELAPSE PREVENTION; maintain even if patient relapses; NOT for elderly
- Disulfiram = DETERRENT (not relapse prevention); CI: cardiac failure, epilepsy, active alcohol, liver disease
- Naltrexone = relapse prevention; CI with opioid agonists (tramadol); NOT in cardiac failure (use acamprosate)

SECTION 7: CLINICAL PHARMACOLOGY & SIDE EFFECTS

7.1 Antipsychotic Side Effects (EPSEs)

EPSE	Time of Onset	Treatment	Key Facts
Acute dystonia	Minutes to hours	Anticholinergics (procyclidine, benztropine)	Young males; high-potency FGA
Pseudo-parkinsonism	Days to weeks	Reduce dose; switch to SGA; anticholinergics	Symmetrical in drug-induced; asymmetrical in PD

EPSE	Time of Onset	Treatment	Key Facts
Akathisia	Hours to weeks	Propranolol (first-line); mirtazapine/mianserin; cyproheptadine; benzodiazepines	Inner restlessness; not relieved by movement alone
Tardive dyskinesia (TD)	Months to years	STOP anticholinergic FIRST; reduce antipsychotic; switch to clozapine/quetiapine; tetrabenazine	Elderly women; lip-smacking, choreiform; IRREVERSIBLE potentially

TD management: First step = STOP procyclidine (anticholinergics worsen TD). Then reduce antipsychotic dose, switch to atypical (clozapine/quetiapine have best evidence). Tetrabenazine = only UK-licensed treatment.

7.2 Mood Stabiliser Key Facts

- Lithium hypercalcaemia: Lithium RAISES calcium (carbamazepine, valproate, haloperidol LOWER it)
- Lithium → hypothyroidism (20% risk in middle-aged women); also hyperthyroidism, hyperparathyroidism
- Lithium toxicity signs (>1.2 mmol/L): coarse tremor, myoclonic jerks, ataxia, confusion, seizures
- Valproate → pancreatitis (epigastric pain radiating to back + vomiting); teratogenic; NOT for women <50 years
- BALANCE study: Lithium $>$ valproate for bipolar prophylaxis; Li+valproate combo $>$ valproate alone
- Bipolar depression treatment options: 1) Olanzapine+fluoxetine, 2) Quetiapine, 3) Olanzapine, 4) Lamotrigine/Lurasidone
- Anticonvulsant discontinuation (as mood stabilisers): minimum 1 month gradual taper

7.3 Clozapine

- Augmentation for partial response: Amisulpride (best evidence), then Aripiprazole
- First step when clozapine not working: CHECK COMPLIANCE (clozapine level)
- Clozapine sedation → worsens motivation (indirect effect); but clozapine directly IMPROVES motivation
- Clozapine CI: paralytic ileus, agranulocytosis history, angle-closure glaucoma (use with caution)
- Weight gain on antipsychotics: Metformin = drug of choice; GLP-1 agonists emerging
- CUTLASS study: FGA = SGA in efficacy and quality of life (except clozapine)

7.4 Antidepressants – Special Situations

Situation	Drug Choice	Drug to Avoid
Post-MI depression	Sertraline (safest); mirtazapine	Fluvoxamine, citalopram (cardiac risk); TCAs
Epilepsy + depression	Mirtazapine, citalopram/escitalopram, sertraline	Bupropion, clomipramine, amitriptyline (proconvulsant)

Situation	Drug Choice	Drug to Avoid
Epilepsy + hyponatraemia (carbamazepine)	Mirtazapine (low risk of hyponatraemia)	SSRIs (high risk of hyponatraemia)
Liver disease	Mirtazapine; imipramine	Avoid valproate, lamotrigine
Renal impairment (eGFR <25)	Haloperidol, olanzapine	Amisulpride, sulpiride (renally cleared)
Taking Tamoxifen	Sertraline, citalopram, venlafaxine	Paroxetine, fluoxetine, duloxetine, bupropion (CYP2D6 inhibitors)
Haemorrhagic stroke + depression	Sertraline or mirtazapine	Avoid antiplatelet SSRIs if active bleeding risk
Breastfeeding	Sertraline (preferred); paroxetine (low milk/plasma)	Lithium (contraindicated), doxepin
SSRI + TCA overdose	Features: hyperthermia, hyperreflexia, clonus = SEROTONIN SYNDROME	Not NMS (NMS = rigidity, slower onset, no myoclonus)

ECT and dose of antidepressants: Lofepramine, Venlafaxine, Moclobemide → reduce/stop before ECT (increase seizure risk or interaction). Benzodiazepines reduce seizure duration (avoid). Lithium → generally continued but ×12 risk of delirium.

7.5 QTc Prolongation

- Risk factors: female, hypokalaemia, hypocalcaemia, hypomagnesaemia, liver disease, congenital long QT, bradycardia, starvation
- QTc >500 ms → reduce dose IMMEDIATELY + consider urgent cardiology referral
- Risperidone + citalopram → QTc prolongation (high-risk combination)
- Citalopram + diphenhydramine → QTc prolongation
- Aripiprazole and olanzapine have least QTc effect

SECTION 8: PSYCHOTHERAPY

8.1 Cognitive Analytic Therapy (CAT) – Very frequently tested

- Initial phase: exploration of TRAPS, DILEMMAS, and SNAGS; therapist writes reformulation letter
- Middle phase: working through problems using diagrams of target problem procedures
- Ending phase: both patient and therapist write GOODBYE LETTERS
- Three Rs: Reformulation → Recognition → Revision
- Sequential diagrammatic reformulation (visual flow chart) = CAT technique

8.2 DBT – Dialectical Behaviour Therapy

- Developed by Marsha Linehan (1991) for BPD / self-harm
- Pre-treatment stage: motivation and commitment to therapy; agreement on goals
- Hierarchy: self-harming behaviours → therapy-interfering behaviours → quality of life
- Components: individual therapy, group skills training, PHONE CALLS (core technique), therapist consultation
- NOT based on psychodynamic formulations; not only done in groups; has hierarchy of goals; NOT a contraindication: low frustration tolerance

8.3 Family Therapy – Distinguish the Models

Model	Key Developer	Key Concepts	Exam Clues
Structural	Minuchin	Clear rules, hierarchy, subsystems, boundaries, coalitions. Therapist = EXPERT stance	Unspoken/unsaid rules; structural hierarchy
Strategic	Haley, Madanes	Distorted hierarchies; vicious cycles; PARADOXICAL INJUNCTIONS; task/goal setting	Paradoxical interventions; strategic stance
Systemic (Milan)	Selvini-Palazzoli	Hypothesising, neutrality, positive connotation, CIRCULAR/INTERVENTIVE QUESTIONING, reflecting teams, RE-FRAMING. NON-EXPERT stance	Re-framing + hypothesising = Milan/systemic
Narrative	White, Epston	Problem-saturated narratives; EXTERNALISATION; unique outcomes. Therapist highlights exceptions	Externalisation + unique outcomes
Psychodynamic	Object relations	Past experiences shape family dynamics; projective identification, scapegoating, splitting, holding environment	Focus on past experiences / object relations
Transgenerational	Bowen	Multi-generational patterns; differentiation scale; triangles; emotional cut-off	Family patterns across generations

The most tested distinction: Systemic = re-framing + hypothesising (Milan model). Strategic = paradoxical injunctions. Structural = unspoken rules + hierarchy. Narrative = externalisation.

8.4 Other Psychotherapies

- Motivational Interviewing (DEARS): Developing discrepancy; Empathy; Avoid arguments; Rolling with resistance; Support self-efficacy. Amplifies cognitive dissonance.
- IPT focus areas (RIIG): Role transitions; Interpersonal deficits; Interpersonal disputes; Grief
- Transference: Is NOT only related to the therapist; NOT always takes months; CAN be psychotic in nature; IS a common phenomenon; explained by PROJECTION (not sublimation)

- Defence mechanisms: Primitive (Level 1) = Denial, Distortion, Splitting, Projection. Neurotic = Displacement, Reaction formation, Repression, Intellectualisation, Dissociation. Mature = Altruism, Sublimation, Suppression, Humour, Identification
- Yalom's curative factors: Catharsis; Altruism; Group cohesion; Installation of hope; Universality (sharing similar experiences); Interpersonal learning
- CBT in psychosis: main role = DISTRESS REDUCTION (not confrontation)
- Behavioural activation assessment: degree of AVOIDANCE (TRAP = Trigger, Response, Avoidance Pattern)

SECTION 9: PREVENTION – SAME QUESTIONS EVERY SESSION

These questions appear in every single paper. The same scenarios are recycled.

Prevention Type	Definition	Example from Recalls
Universal (Primary)	Targeted to the whole population; no risk criteria needed	School-wide anti-bullying programme; school smoking programme; antibullying retreat
Selective (Primary)	Targeted to HIGH-RISK subgroups (biological/psychological/social risk factors)	Low-income families mental health support; children of parents with mental illness
Indicated (Primary)	Targeted to individuals with DETECTABLE early signs but below diagnostic threshold	At-risk mental state (ARMS/UHR) + CBT for psychosis prevention
Secondary Prevention	Early detection and treatment of established disease before symptoms	Screening programmes
Tertiary Prevention	Care of ESTABLISHED disease; minimise negative effects and prevent complications	Lithium to prevent further bipolar episodes

ARMS/at-risk mental state = BOTH primary AND indicated prevention. Always choose TWO answers when asked this.

Universal prevention for alcohol in the community = increase price per unit (most evidence). Education in schools = universal too.

SECTION 10: FORENSIC PSYCHIATRY

10.1 Suicide in Prison

- Rate: 3–8 times higher than general population in males; >10 times higher in females
- Highest risk group: remand prisoners (not yet sentenced), young offenders, life sentence prisoners, those with substance misuse history

- Sentenced status = LOWER risk than remand status
- Occupying a SINGLE CELL = strong risk factor (not protective)
- Being MARRIED in prison = RISK factor (opposite of outside prison)
- Most common method: hanging
- Most suicides within 6 months of imprisonment
- Most are in MEN (>90%), but women have proportionally higher excess mortality

10.2 Fitness to Plead – Pritchard Criteria

- Must understand the charges
- Must be able to instruct solicitor/counsel
- Must be able to decide plea (guilty/not guilty)
- Must be able to follow court proceedings
- Must be able to challenge jurors
- Must be able to give evidence in own defence

AMNESIA for the offence does NOT make someone unfit to plead. Most common condition causing unfitness to plead = learning disability.

10.3 Automatism

- Sane automatism: EXTERNAL cause (concussion, dissociation, single episode hypoglycaemia, sleep terrors) → complete acquittal
- Insane automatism: INTERNAL cause (epilepsy, sleepwalking, brain tumour, recurring diabetes) → not guilty by reason of insanity

Sleep terrors = SANE automatism (extrinsic). Epilepsy = INSANE automatism (intrinsic).

10.4 Risk Factors for Reoffending

- Static factors (fixed, historical): age at first offence, gender, previous offences, index offence severity
- Dynamic factors (changeable, short-term relevant): substance misuse, lack of employment, attitude to offending, anger expression, engagement with treatment
- Sexual recidivism highest in: DEVIANT SEXUAL INTERESTS + ANTISOCIAL orientation (strongest combination)
- Indecent exposure: 20–30% reoffend; 80% new offenders; most stop after first conviction
- ASPD prevalence in prison = 50%; homicide perpetrators with schizophrenia = 5%
- Arson: most common psychiatric comorbidity = substance use disorder (especially alcohol) + ASPD

10.5 Key Legal Cases

Case	Significance
Bournewood	Deprivation of Liberty Safeguards (DoLS); led to MCA amendments
Tarasoff	Duty to warn third parties of patient's violent intentions

Case	Significance
Osman v UK	Police have duty to warn potential victims when they have relevant information
Gillick / Fraser Guidelines	Treatment of minors without parental consent
Pritchard	Criteria for fitness to plead
Bolam	Standard of medical negligence

10.6 Homicide

- Female perpetrators of homicide: most victims are <16 years of age (especially if perpetrator has psychosis)
- Filicide-suicide (parent kills child then themselves): 13% of filicide cases
- Homicide-suicide (domestic): most commonly associated with DEPRESSION (not schizophrenia)
- Schizophrenia accounts for 5% of all homicide perpetrators
- Diminished responsibility applies to MURDER charges (reduces to manslaughter)
- Infanticide: most common MH diagnosis = AFFECTIVE disorder (depression/psychosis)

SECTION 11: ECT

- Most common memory affected: RETROGRADE (explicit/episodic); also anterograde but retrograde > anterograde
- Absolute CI: pheochromocytoma ONLY
- Bilateral ECT: more effective than unilateral BUT more cognitive side effects
- Right unilateral placement: less cognitive impairment but needs higher dose for efficacy
- Remission rate in treatment-resistant depression: 60–80% (up to 90% in psychotic depression)
- Relapse after ECT without maintenance treatment: >80% within 6 months
- ECT vs rTMS: ECT is MORE effective; rTMS is more TARGETED/specific area; ECT needs MORE FREQUENT sessions
- ECT + lithium: x12 risk of delirium (avoid combination if possible)
- NICE indications: severe/life-threatening depression; treatment-resistant depression; severe mania; catatonia; psychotic depression
- Best predictor of ECT response: PSYCHOTIC DEPRESSION (not severity of depression or melancholic features)

Drug	Effect on Seizure Duration
Benzodiazepines	REDUCE (avoid if possible)
Anticonvulsants (as mood stabilisers)	REDUCE (continue but prepare for higher energy)
Clozapine, phenothiazines	MAY INCREASE

Drug	Effect on Seizure Duration
SSRIs, MAOIs, venlafaxine	Minimal effect
Lithium	Possibly increased; x12 delirium risk
TCAs	Possibly increased; arrhythmia risk in elderly

SECTION 12: EATING DISORDERS

Feature	Anorexia Nervosa	Bulimia Nervosa
Prevalence	0.5–1%	~2% (MORE common than AN)
Social class	All classes equally	All classes equally
Impulse control	INTACT	POOR (shoplifting, substance misuse)
Weight	Low (BMI <17.5)	Usually normal range
Lanugo	Present	Absent
Laxative abuse	Can occur	More common
First-line treatment (adults)	FT-AN (family therapy)	Guided self-help (CBT-based)
First-line treatment (adolescents)	FT-AN (family therapy)	Bulimia-focused family therapy (FT-BN)
Pharmacotherapy	No medication as sole treatment	Fluoxetine (60mg) licensed; guided self-help first

- Refeeding syndrome: most dangerous electrolyte disturbance = HYPOPHOSPHATEMIA (phosphate drops as carbohydrate intake drives insulin release and intracellular phosphate shift)
- Other refeeding disturbances: hypokalaemia, hypomagnesaemia, occasionally hypocalcaemia
- MEED red flags requiring admission: HR <40, BMI <13 (adult), temp <35.5°C, QTc >430ms (males) / >450ms (females), hypokalaemia <2.5, significant dehydration

SECTION 13: CHILD & ADOLESCENT PSYCHIATRY

13.1 ADHD

- Heritability: 70–80% (highest of all child psychiatric conditions listed)
- Age criterion ICD-11 and DSM-5: symptoms present BEFORE age 12
- Stimulants first-line (methylphenidate); atomoxetine for tics as SIDE EFFECT of stimulant; clonidine/guanfacine for COMORBID tic disorder
- ADHD + liver disease (child who wants less sedation) → Guanfacine

- If ADHD + aggressive behaviour, not responding to stimulants/atomoxetine → Risperidone
- Pathophysiology: frontostriatal pathway

13.2 OCD in Children

- Sertraline: licensed from age 6 years for paediatric OCD
- Fluvoxamine: licensed from age 8 years
- First-line drug (when sertraline not in options): fluvoxamine
- Clomipramine: NOT first-line (cardiac monitoring needed); used in refractory cases
- Paroxetine: NOT recommended in children
- NICE: SSRIs before clomipramine; clomipramine + SSRI combination = refractory OCD (Clomipramine + Citalopram)
- Refractory OCD surgery: Anterior cingulotomy (if not available → DBS, but still research only)

13.3 Depression in Children

Severity	Age	First-line Treatment
Mild (watchful waiting)	All ages	Watchful waiting 2 WEEKS (NOT 6 weeks; NOT 2 months), then digital CBT/group CBT/group IPT
Mild depression (12-year-old)	12 years old	Family therapy, CAT, or digital CBT (NOT DBT)
Moderate-severe (5–11 years)	5–11	Family-based IPT, family therapy, psychodynamic psychotherapy, individual CBT ± fluoxetine
Moderate-severe (12–18 years)	12–18	Individual CBT ± fluoxetine
Treatment-resistant / psychotic	Any age	Intensive psychological therapy ± fluoxetine/sertraline/citalopram ± antipsychotic
Minimum age for antidepressant	5 years	Fluoxetine from age 5 years (NICE)

13.4 Tic Disorders

- Transient tic disorder: tics < 1 year duration
- Chronic motor OR chronic vocal tic: same type of tic > 1 year
- Tourette syndrome: 2+ motor tics AND 1+ vocal tic for > 1 year; MALE:FEMALE = 2–5:1
- Simple motor tics: head shaking, eye blinking, grimacing, shoulder shrugging
- Simple vocal tics: throat clearing, grunting, barking, coughing
- Copropraxia: involuntary obscene gestures

13.5 Autism Spectrum Disorder

- Now and next cards: show schedule of planned activities in pictures/symbols

- Social stories (Carol Gray): short personalised stories helping understand social situations; used for predictable visits (e.g. mother visiting)
- Intensive interaction: body-language based, mirrors gestures and vocalisations
- Makaton: sign language for communication
- ICD-11 ASD: must show developmental difficulties BEFORE age 3
- Risk factors: genetic loading (strongest); low 5-min APGAR score; various perinatal factors

SECTION 14: ICD-11 PERSONALITY DISORDERS

The ICD-11 PD classification is heavily tested. There are 5 trait domains. Know the match between domain and old PD categories.

ICD-11 Trait Domain	Core Feature	Corresponds to Old PD
Negative Affectivity	Broad range of negative emotions; emotional lability; low self-esteem; mistrustfulness	Borderline (emotional lability); Paranoid (mistrustfulness)
Detachment	Social detachment (avoidance of interactions); emotional detachment (aloofness, limited expression)	Schizoid; Schizotypal
Dissociality	Disregard for others' rights; self-centredness; lack of empathy; manipulateness; callousness	Antisocial / Dissocial; Narcissistic
Disinhibition	Impulsivity; recklessness; irresponsibility; lack of planning; distractibility	Borderline (impulsive type); Antisocial
Anankastia	Perfectionism; rigid adherence to rules; emotional constraint; stubbornness; risk-avoidance; perseveration	OCPD / Anankastic

Exam scenario: Man fired multiple times, feels he has enough skills, impulsively books holiday that morning = DISINHIBITION domain.

Borderline PD in ICD-11 = Personality disorder with Negative Affectivity + Disinhibition qualifiers, with 'borderline' as a specifier.

SECTION 15: OTHER HIGH-YIELD CLINICAL TOPICS

15.1 Self-Harm & Suicide Statistics

- Suicide within 1 year of self-harm: 0.7% (1 in 100); males 1.1%, females 0.5%
- This is 66 times the general population rate

- Within 3 months of discharge from inpatient: 25% of suicides
- 40% of suicides have visited GP in the week before
- Lifetime prevalence of suicide: 4%; lifetime suicidal thoughts: 15%
- Lifetime suicide rate in EUPD (BPD): approximately 10%
- BPD remission at 10 years: ~50% (some sources: 30–50%)

15.2 ECG Findings in Psychiatry (EMI format tested every session)

ECG Finding	Condition / Drug
QTc prolongation (>500ms)	Citalopram + diphenhydramine; risperidone + sertraline; clozapine; quetiapine
Saddle-shaped ST elevation (T wave changes)	Clozapine myocarditis (chest pain + troponin)
Sinus bradycardia	Donepezil (ankle oedema, fatigue)
Hypokalaemia changes (ST depression + U wave)	Eating disorder (purging, laxative abuse)
Low voltage QRS	Hypothyroidism; cardiac tamponade
Prolonged PR interval	Anorexia with electrolyte disturbance

15.3 Dysphasia / Aphasia Types

Type	Fluency	Comprehension	Repetition	Location
Broca's (non-fluent expressive)	Impaired	Intact	Impaired	Left frontal (Broca's area)
Wernicke's (fluent)	Intact (jargon)	Impaired	Impaired	Left temporal
Conduction	Intact	Intact	IMPAIRED	Arcuate fasciculus
Global	Impaired	Impaired	Impaired	Large left hemisphere lesion
nfvPPA (FTD)	Non-fluent, slow, effortful	Intact single words	Impaired	Left frontal operculum
svPPA (semantic)	Fluent	Impaired single words	Intact	Left anterior temporal
Transcortical motor	Impaired	Intact	INTACT	Left frontal (supplementary motor)

15.4 CBT Cognitive Distortions (tested as EMI)

Distortion	Definition	Exam Scenario
Arbitrary inference	Drawing conclusion without evidence (mind reading)	'I saw her talking to him, she must be cheating'
Overgeneralisation	One failure → all failures	'I failed once so I'll always fail'
Catastrophisation	Trivial event → disaster (sequence)	Palpitations → heart attack → death
Dichotomous thinking (all-or-nothing)	No middle ground	'I had one slice of cake, my diet is ruined'
Selective abstraction (filtering)	Focus only on negative, ignore positives	Got one bad comment in good review → focuses on that
Minimisation	Downplay positives	'He praised my slide only because he had to' = DISCOUNTING POSITIVES
Magnification	Exaggerate negatives	'I made a small mistake, I am a complete failure'
Personalisation	External events attributed to self	'Friend cancelled because of me'
Emotional reasoning	Feelings = facts	'I feel like a burden therefore I am'
Reaction formation	Acting opposite to true feelings	Grandmother nourishes child but unconsciously despises her

15.5 Stages of Change (Prochaska & DiClemente)

- Pre-contemplation: no intention to change ('doesn't want to stop smoking')
- Contemplation: considering change but ambivalent
- Preparation: planning to change in next month ('planning to stop smoking in a month')
- Action: actively making change
- Maintenance: sustaining change (e.g., 4 years within 14 units/week = MAINTENANCE)
- Relapse: returning to old behaviour ('heroin again after a week of abstinence')

15.6 Equal M:F Prevalence – Frequently Tested

- Equal prevalence (M=F): OCD (1:1); Hypochondriasis; GAD; Bipolar type I; Prader-Willi; Fragile X (expressed differently)
- More common in females: Depression (1:2); BPD (1:3); Bulimia (1:9); Anorexia (1:9); Panic (1:2–3)
- More common in males: Autism (3:1); ADHD (2–5:1); Tourette's (2–5:1); Antisocial PD (5:1); Asperger's (5:1)

15.7 Key Epidemiological Facts

- ADHD heritability = highest of childhood psychiatric conditions (70–80%)
- ASD heritability = ~90% (genetic loading is the single strongest risk factor)
- Schizophrenia heritability = 80%; lifetime risk general population = 1%; identical twin = 48%

- Most common modifiable risk factor for late-life dementia (population attributable risk) = SMOKING
- Postnatal depression prevalence = 10–15% of new mothers
- ASPD in prison population = 50%
- International prison suicide rates: higher than general population; most are male
- LD populations die ~20 years earlier than general population
- OCD prevalence in adults = 1–3%; M:F = 1:1 in adults; peaks age 10–12 in children

SECTION 16: HOW TO CONCENTRATE YOUR REVISION

Based on frequency analysis across all 8 sources, here is exactly where to focus your time:

Priority	Topic	Time Allocation	Key Resource
1 (CRITICAL)	Statistics & Research Methods	35% of revision time	SPMM mocks; Fresh Questions stats section
2 (CRITICAL)	LD syndromes & genetics table	10% – memorise once, revisit weekly	One comprehensive table; SPMM videos
3 (HIGH)	Drugs in pregnancy & neonatal effects	8%	SPMM + Maudsley guidelines
4 (HIGH)	Dementia (esp. DLB, Alzheimer's, FTD)	8%	Oxford Handbook OA Psychiatry
5 (HIGH)	Pharmacology (EPSEs, clozapine, mood stabilisers)	8%	Maudsley prescribing guidelines
6 (HIGH)	Psychotherapy (CAT, DBT, family therapy)	6%	SPMM psychotherapy module
7 (MEDIUM)	Forensic psychiatry	5%	SPMM forensic module
8 (MEDIUM)	Eating disorders	5%	NICE guidelines
9 (MEDIUM)	Child psychiatry / CAMHS	5%	NICE guidelines + SPMM
10 (MEDIUM)	Prevention	3% – same 5 questions	Memorise the 3+3 table

16.1 SPMM-Specific Preparation

- Mock 14 (April 2024) was directly referenced in March 2024 exam recalls – do this mock in full
- Fresh Questions 2023 and 2022 are referenced in Oct 2023 and Sep 2022 recalls – do these sets
- Psychmentor Q&A bank covers many recall questions – use it alongside SPMM

- The stats EMI stems with forest plots, Kaplan-Meier curves, and CONSORT diagrams are NEW each session but test the SAME skills – practice interpretation not memorisation
- SPMM videos: watch the stats, psychotherapy, and LD syndrome videos until you can reproduce key tables from memory

16.2 The Golden Rules

Never skip stats. There are 40-50 stats questions per paper. Getting most of them right is the difference between passing and failing.

The same LD syndrome questions come up every session. Learn ONE good table covering all 12 syndromes and you will get most of these marks.

Prevention questions are the easiest marks in the exam. The same 5 scenarios recycle every session. Memorise the definitions once and bank these 3-4 marks every time.

For psychotherapy: the distinction between CAT (3 Rs, goodbye letters), DBT (phone calls, hierarchy), and family therapy types (structural/strategic/systemic) is tested in every paper.

For ECT memory: Retrograde (explicit) memory is most affected. Bilateral > unilateral efficacy but more cognitive SEs. QTc prolongation: >500ms = emergency reduction.

Good luck with your MRCPsych Paper B examination!

Based on systematic analysis of 6 recall sessions (Sep 2022 – May 2025) + SPMM Mock 14 + compiled recalls